A fixed interval already excludes square-root growth

Disproof of the printed asymptotic in TLMC 00000001906

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Abstract

For the conventional unweighted, rough Rubio de Francia square function on the real line, every admissible family of constants satisfies $C_p \geq p/(8\pi e)$ for all real $p \geq 2$. A single fixed frequency interval suffices. The Lean proof constructs the native Fourier projections of genuine interval indicators, proves their integral identity by Fubini, and derives the native L^p bounds. It excludes even $C_p = O(\sqrt{p})$, hence any finite-coefficient square-root asymptotic.

1 Precise model and scope

The printed bilingual conjecture names the Rubio de Francia inequality for arbitrary intervals and proposes a definite-coefficient $p^{1/2}$ asymptotic. It does not specify the underlying domain or a Fourier normalization. We use the usual real-line, unweighted, *rough* interval projections, as in [1, pp. 1–2] and [2, p. 2, Eq. (1.1)]:

$$\mathcal{F}f(\xi) = \int_{\mathbb{R}} f(y)e^{-2\pi iy\xi}\,dy, \qquad P_{[a,b]}f = \mathcal{F}^{-1}(\mathbf{1}_{[a,b]}\mathcal{F}f), \qquad S_\Omega f = \Big(\sum_{I\in\Omega} |P_I f|^2\Big)^{1/2}.$$

Here Ω is any finite family of pairwise disjoint bounded intervals. Endpoints have Lebesgue measure zero. A nonnegative C is called admissible at p when the native extended L^p inequality $\|S_\Omega f\|_p \leq C\,\|f\|_p$ holds for all these families and all integrable $f \in L^p(\mathbb{R})$. Every usual Rubio de Francia constant necessarily satisfies this test class. The singleton family $\{[0,1]\}$ has $S_\Omega f = |P_{[0,1]}f|$.

We refute only the printed square-root asymptotic under this standard model. We neither dispute Rubio de Francia boundedness nor compute a sharp linear coefficient. We do not replace the operator by a smooth, martingale, or periodic square function. Theorem 1.1 of [1] concerns the lacunary operator; its later discussion separately treats arbitrary intervals on the circle. Those results motivate the scope but are not assumptions of our Lean proof. No transference or dilation theorem is needed.

2 The actual Fourier projection

For $R \geq 1$, put $f_R = \mathbf{1}_{[-R,R]}$. Native Lebesgue integration proves $f_R \in L^1 \cap L^p$, $p > 0$, and

$$\|f_R\|_p = (2R)^{1/p}.$$

The Fourier transform is bounded and continuous. Its restriction to $[0,1]$ is integrable, so its native inverse Fourier transform is continuous. The product phase in the iterated frequency/spatial integral has modulus one on a finite product measure. Fubini and the elementary exponential integral therefore give, whenever $x > R$,

$$P_{[0,1]}f_R(x) = \int_{-R}^{R} K(x-y)\,dy, \qquad K(t) = \frac{e^{2\pi it}-1}{2\pi it} \quad (t>0).$$

Every kernel argument here is strictly positive. Thus no totalized division at zero is used. Substitution $x = R+\delta$, $\delta > 0$, and taking the imaginary part yield the proved identity

$$\operatorname{Im} P_{[0,1]}f_R(R+\delta) = \frac{1}{2\pi}\int_{\delta}^{2R+\delta} \frac{1-\cos(2\pi t)}{t}\,dt. \tag{1}$$

The explicit integral is derived from the actual native Fourier operator; it is not substituted for its definition.

3 A uniform oscillatory bound

For $1 \le a \le b$, differentiate

$$F(t) = \frac{\sin(2\pi t)}{2\pi t}, \qquad F'(t) = \frac{\cos(2\pi t)}{t} - \frac{\sin(2\pi t)}{2\pi t^2}.$$

The fundamental theorem of calculus, $-1 \le \sin \le 1$, and the actual integral of t^{-2} give

$$\int_a^b \frac{\cos(2\pi t)}{t}\, dt \le \frac{1}{2\pi b} + \frac{1}{2\pi a} + \frac{1}{2\pi}\Big(\frac{1}{a} - \frac{1}{b}\Big) = \frac{1}{\pi a} \le 1.$$

Only this one-sided bound is needed. If $1 \le \delta \le 2$, then $(2R + \delta)/\delta \ge R$. Combining the logarithmic integral with (1) proves

$$\operatorname{Im} P_{[0,1]} f_R(R + \delta) \ge \frac{\log R - 1}{2\pi}. \tag{2}$$

4 Native norm bounds and the asymptotic contradiction

Take any real $p \ge 2$ and $R = e^p$. The genuine interval $E_p = [R + 1, R + 2]$ has Lebesgue measure one. On this entire interval, (2) and $\operatorname{Im} z \le |z|$ imply

$$|P_{[0,1]} f_R(x)| \ge \frac{p - 1}{2\pi}, \qquad \big\|P_{[0,1]} f_R\big\|_p \ge \frac{p - 1}{2\pi}, \qquad \|f_R\|_p = 2^{1/p} e \le 2e.$$

The proof of the output lower bound uses monotonicity of the native `eLpNorm` and the norm of a constant indicator on the actual measure-one interval. If a finite admissible bound is assumed, continuity and that bound explicitly imply native `MemLp` for the output. No infinite norm is converted to a default real zero.

Applying the true singleton inequality now gives

$$\frac{p - 1}{2\pi} \le 2eC_p, \qquad C_p \ge \frac{p - 1}{4\pi e} \ge \frac{p}{8\pi e}.$$

Theorem 1. *For every family $C : \mathbb{R} \to \mathbb{R}$ admissible at every real $p \ge 2$, native $C \ne O_{p\to\infty}(\sqrt{p})$. Moreover, for every finite real L, the actual ratio $C_p/\sqrt{p}$ does not converge to L.*

Proof. A supposed eventual bound $|C_p| \le K\sqrt{p}$ contradicts the linear lower bound by taking $p = t^2$ with t larger than both the threshold and $8\pi eK$. A convergent ratio is eventually bounded, which would give precisely such a native big-O bound. Both conclusions are proved for the real `atTop` filter, so no passage from integer exponents is assumed. □

No optimal constant is defined by an unchecked infimum, and no norm, moment, or growth value is assigned.

5 Formal coverage and reproduction

The main namespace is `TLMC1906`. The Fourier identities are proved in:

`projection_input_double`, `projection_input_kernel_shift`, `projection_input_im`.

The actual integral and norm estimates are proved in:

`cosine_integral_bound`, `exponential_output_lower`, `Admissible.output_memLp`, `Admissible.linear_lower`.

The native asymptotic conclusions are:

`no_sqrt_bigO`, `no_finite_sqrt_coefficient`.

The project pins Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d`. From `lean/`, run:

```
lean --version
lake build +Main
lake env lean -DwarningAsError=true Check.lean
lake env leanchecker --fresh Main
```

Equivalently, run `lean -version` followed by `./verify.sh` for the remaining three commands. The exhaustive check traverses all owned declarations, including generated ones, and invokes ordinary `#print axioms`. Only `propext`, `Classical.choice`, and `Quot.sound` are permitted. Source-bound command, cleanup, hash and CPU-affinity receipts accompany the submission. Production timings are diagnostic observations, not controlled efficiency evidence. The complete original bilingual statement is retained in `SOURCE.md`.

References

[1] R. Garg, L. Roncal and S. Shrivastava, *Quantitative weighted estimates for Rubio de Francia's Littlewood–Paley square function*, arXiv:1809.02937v2 (7 October 2019), pp. 1–2. https://arxiv.org/abs/1809.02937.

[2] F. Di Plinio, M. Flórez-Amatriain, I. Parissis and L. Roncal, *Pointwise localization and sharp weighted bounds for Rubio de Francia square functions*, arXiv:2308.01442v3 (2024), pp. 1–2, especially Eq. (1.1). https://arxiv.org/abs/2308.01442.